

NANYANG JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 2

MATHEMATICS

9758/01

Paper 1

10th September 2018

3 Hours

Additional Materials: Answer Paper
List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

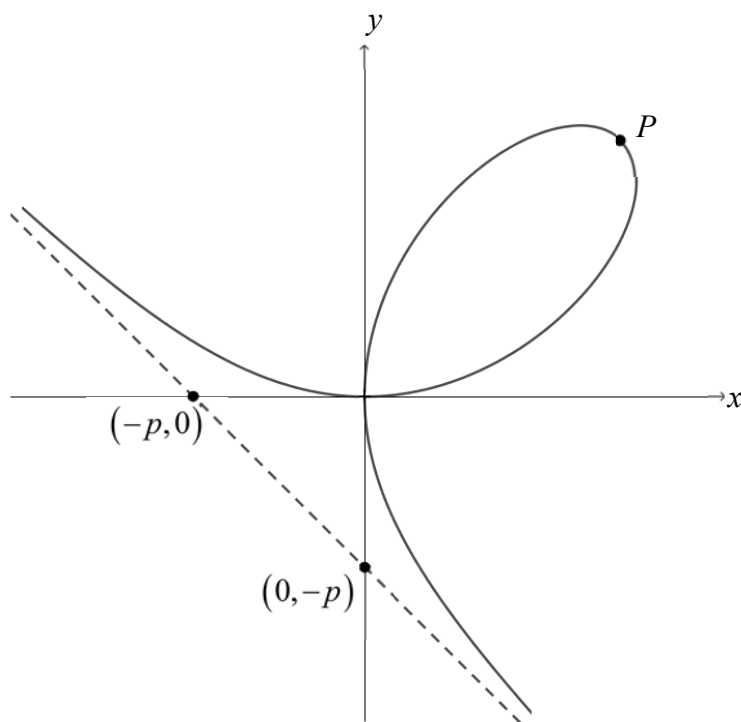
The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **6** printed pages.



NANYANG JUNIOR COLLEGE
Internal Examinations

- 1 A departmental store sells a pair of jeans at \$46.90, blouses at \$29.00 each and a pair of shoes at \$19.90. Items that are priced at more than \$25 are sold at a further discount of 15%. Betty bought twice as many blouses as jeans and she charged \$257.93 to her credit card for a total of 10 items. How many pairs of jeans, shoes and blouses did she buy? [4]
- 2 The diagram below shows the folium of Descartes curve with equation $x^3 + y^3 = 3pxy$, where $p > 0$, and the asymptote of the curve passes through the points $(-p, 0)$ and $(0, -p)$.



Point P lies on the curve such that the tangent at P is parallel to the asymptote of the curve. Find the coordinates of point P in terms of p . [6]

- 3 A curve C has equation $y^2 + 14y + 4x^2 + 16x + 16xy + 13 = 0$.
- (i) If a real value of x is substituted into the equation, it becomes a quadratic equation in y . Given that there are two distinct values of y for this equation, show that $5x^2 + 8x + 3 > 0$, and hence find the set of possible values of x . [5]
- (ii) Find the coordinates of the points where C cuts the y -axis. State with a reason whether C is a graph of a function. [2]

4 A curve D has equation $y = x|x - a|$, $a > 0$.

(i) Describe a pair of transformations which transforms the graph of D on to the graph of $y = 2 - x|x + a|$. [2]

(ii) Sketch D , giving the coordinates of the axial intercepts and turning point in terms of a . [2]

(iii) On a separate diagram, sketch the curve $y = \frac{1}{x|x - a|}$, giving the coordinates of the turning point and the equations of the asymptotes in terms of a . [3]

(iv) State the range of values of k if $\frac{1}{x|x - a|} = k$ has exactly one solution. [1]

5 The line l passes through the points A and B with coordinates $(5, 2, 4)$ and $(4, -1, 3)$ respectively. The plane p has equation $4x + 7y + 5z = 24$.

(i) The point C lies on l such that the foot of perpendicular of C onto p has coordinates $(3, 1, 1)$. Find the coordinates of C . [4]

Plane p_1 has equation $3x - 2y + \lambda z = \mu$.

(ii) What can be said about the values of λ and μ if l does not intersect p_1 ? [2]

(iii) Hence find the exact values of μ if the distance between p_1 and l is 2 units. [3]

6 The functions f and g are defined by

$$f : x \mapsto \sin x \cos x, \quad x \in \mathbb{R}, -\frac{\pi}{2} < x \leq \frac{\pi}{2},$$

$$g : x \mapsto \frac{1}{x}, \quad x \in \mathbb{R} \setminus \{0\}, -1 \leq x \leq 1.$$

(i) A function h is said to be odd if $h(-x) = -h(x)$ for all x in the domain of h . Show that g is odd and determine if f is odd. [2]

(ii) Explain why f does not have an inverse. If the domain of f is further restricted to $0 < x \leq a$, where $a \in \mathbb{R}$, the function f^{-1} will exist. State the largest possible exact value of a . [2]

Use the value of a in (ii) for the rest of the question.

(iii) Sketch the graphs of f , f^{-1} and ff^{-1} on the same diagram. [3]

(iv) State with reasons, whether the composite functions fg and gf exist. If the composite function exists, find the rule, domain and range. [3]

- 7 (a) Referred to the origin O , points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are unit vectors.
- (i) By using scalar product, show that the vector $\mathbf{a} + \mathbf{b}$ is the bisector of angle AOB . [3]
- (ii) If the area of the triangle AOB is $\frac{1}{\sqrt{10}}$ units², state the exact value of the sine of angle AOB . [1]
- (b) Referred to the origin O , points C and D have position vectors $7\mathbf{i} - 8\mathbf{j} + 7\mathbf{k}$ and $4\mathbf{i} + 7\mathbf{j} + 4\mathbf{k}$ respectively.
- (i) Using vector product, find the exact shortest distance of the line, passing through points C and D , from the origin. [4]
- (ii) Find angle OCD . [2]

- 8 A curve C is represented by the parametric equations

$$x = t^2(t + 6), \quad y = t^2 + t - 6, \quad \text{for } t < 0.$$

- (i) Find the equation(s) of the tangent to the curve C which is parallel to the y -axis. [3]
- (ii) Sketch C , showing clearly the axial intercepts. [2]
- (iii) Let R be the finite region bounded by C and the line $x = 16$. Find the area of R . [5]

- 9 (a) The sum, S_n , of the first n terms of a sequence $u_1, u_2, u_3, \dots$ is given by

$$S_n = \frac{6}{13} \left(1 - \frac{1}{3^{3n}} \right).$$

- (i) Given that the series $\sum u_r$ converges, find the smallest integer n for which S_n is within 10^{-8} of the sum to infinity. [3]
- (ii) Find a formula for u_n in simplified form. [2]
- (b) Using the formulae for $\sin(A \pm B)$, prove that
- (i) $\sin(2r+1)\theta - \sin(2r-1)\theta = 2 \cos 2r\theta \sin \theta$. [1]
- (ii) Hence find an expression for $\sum_{r=1}^n \sin^2 r\theta$, giving your answer in terms of $\cos(n+1)\theta$, $\sin n\theta$, $\sin \theta$ and n , where $\theta \neq k\pi, k \in \mathbb{Z}$. [5]

- 10** The two blades of a pair of scissors are fastened at the point A . The distance from A to the tip of the blade at point B is m cm. Let the angle formed by the line AB and the bottom edge of the blade BC be α radians and the angle between AB and AC be θ radians (Figure 1). A piece of paper resting at point C is cut. As the paper is being cut, the blades come closer to each other and the length of AC increases as shown in Figure 2.

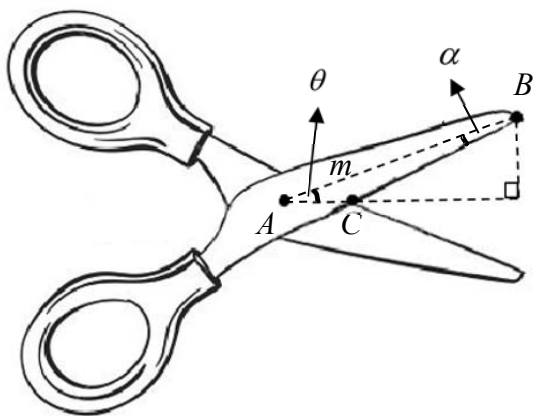


Figure 1:

Scissors when the blades are further apart

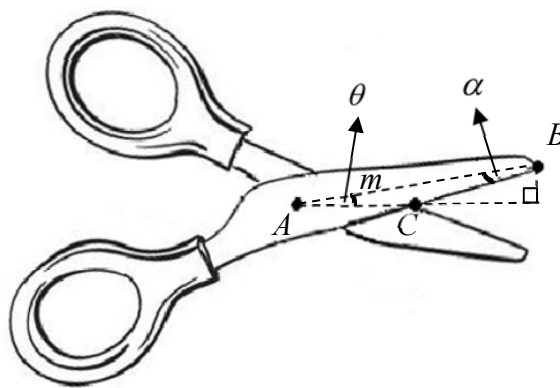


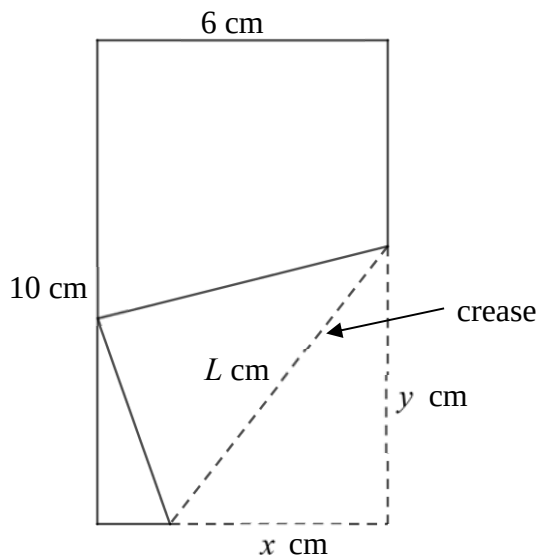
Figure 2:

Scissors when the blades are closer together

- (i) By letting $AC = l$ cm, show that $l = \frac{m \sin \alpha}{\sin(\theta + \alpha)}$. [1]
- (ii) Find $\frac{dl}{d\theta}$ in terms of m , θ and α . [1]
- (iii) It is given that $m = 15$, $\alpha = \frac{\pi}{60}$ and θ is decreasing at a rate of $\frac{5\pi}{9}$ radians per second. Find the rate at which the paper is being cut at the instant when $\theta = \frac{\pi}{9}$. [3]

Question 10 continues on the next page

A triangle is cut out from a rectangular piece of paper measuring 10 cm by 6 cm using the scissors. To form this triangle, the right-hand corner is folded over so as to reach the left-most edge of the paper, forming a crease for the scissors to cut along, as shown in the diagram below. Let the length of the crease be L cm, the base of the triangle to be folded be x cm and the height of the triangle be y cm.



(iv) Show that $L = \sqrt{x^2 + \frac{3x^2}{x-3}}$. [3]

(v) Using differentiation, find the minimum length of the resulting crease. [4]

- 11 A patient in the hospital is being administrated a certain drug through an intravenous (IV) drip at a constant rate of 30 mg per hour. The rate of loss of the drug from the patient's body is proportional to x , where x (in mg) is the amount of drug in the patient's body at time t (in hours). The amount of drug in the patient's body needs to reach 120 mg for the treatment to be effective.

- (i) Explain why the rate of change of x needs to be positive. [1]
- (ii) Initially, there are no traces of the drug in the patient's body, and after 4 hours, the amount of drug in the patient's body is 82.6 mg. Show that

$$x = A \left(1 - e^{-\frac{t}{5}} \right), \text{ where } A \text{ is a constant to be determined.} \quad [5]$$

- (iii) Find the time needed for the amount of drug in the patient's body to reach 120 mg. [1]

A medical worker, who studied mathematical biology, proposed that the rate of change of the amount of drug in the patient's body actually satisfies the differential equation

$$\frac{d^2x}{dt^2} = \frac{1}{\sqrt{2500 - 9t^2}}.$$

- (iv) Find the general solution for the proposed differential equation, given that $x = 0$ when $t = 0$. [6]